

Curious Arduino Box

Problem

Many children's technology toys rely on screens or mobile devices, limiting opportunities for hands-on exploration. The Curious Arduino Box addresses this gap with an interactive, screen-free toy that encourages curiosity, problem-solving, and analysis through direct physical interaction.

Process

The prototype combines an Arduino, switch, servo motors, and LED. When the switch is pressed, programmed commands coordinate the box and arm servos through a sequence of movements, while the LED provides a visual response. The servos then return to their starting positions, allowing the cycle to repeat. The project involved wiring the circuit, programming the motion sequence, and testing the system's behavior.

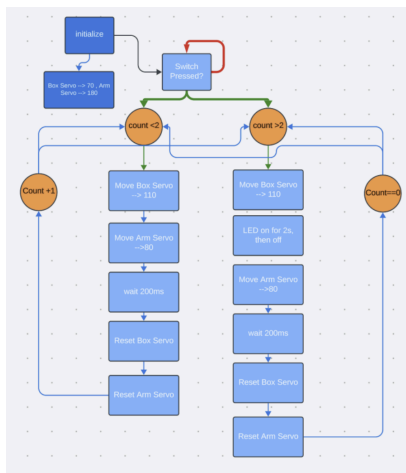


Figure 1: System Logic

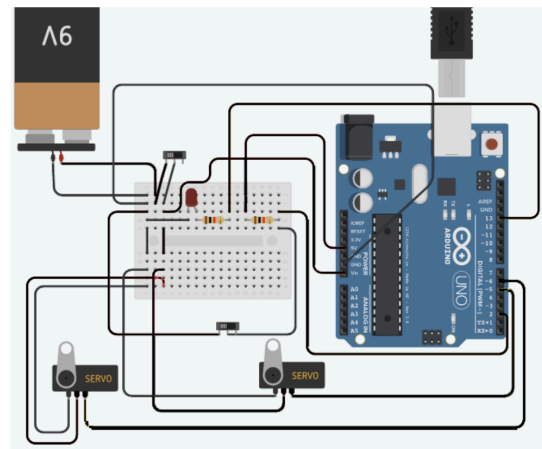


Figure 2: System Wiring Diagram

Product

The final product is a compact interactive box that contains the electronics and moving mechanisms within a simple enclosure. Its features include a moving top, mechanical arm, and LED response, all controlled by the Arduino. Together, these elements create an engaging, screen-free toy that invites children to interact, observe results, and think about how the system works.

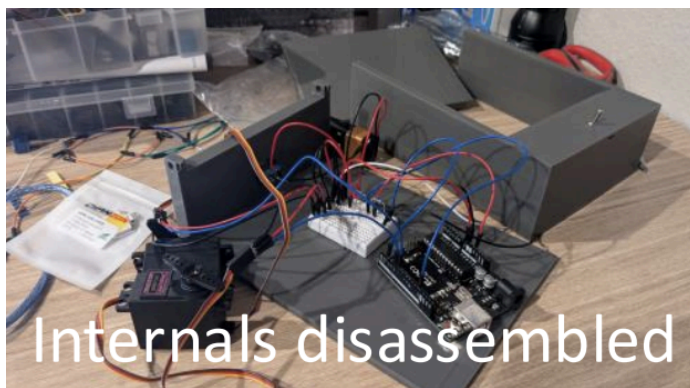


Figure 3: Alpha Prototype Internals



Figure 4: Full Alpha Prototype